

Project 1 — Week 1 stepping stone

DATA 422: Advanced Data Science Methods and Ethics

Due: Wednesday, September 2, 11:59 p.m.
Counts toward: [Project 1](#) (not a lab grade)

Lab 1 practice questions are ungraded. This checkpoint is the start of the wildfire risk model.

What to submit

A short write-up (PDF or notebook) with four pieces.

1. Target contract (one paragraph)

In your own words, state:

- the **unit** (one de-identified structure / DINS inspection);
- the **label serious_damage** (Major or Destroyed = 1; No Damage, Affected, Minor = 0);
- the **conditioning**: the structure was inspected in or near a wildfire perimeter;
- what this number is **not** (not annual household risk, not a premium, not a coverage decision).

2. Privacy and leakage (short list)

Confirm the released table has no street address, APN, or lat/lon.

Name **two** fields you would *not* treat as pre-fire features if they appeared (examples: defensive actions taken during the fire, counts of burned cars). Say why.

3. PyTorch logistic baseline

Fit logistic regression **in PyTorch**. Lab 1 used the book's `Sequential + Sigmoid + BCELoss` on a toy example. This stone uses the numerically stabler wrapper: one linear layer, **logits**, `BCEWithLogitsLoss` (same $\hat{y} = \sigma(z)$).

Use the instructor table and the supplied `split` column. Fit on **train**. Report on **validation** (leave **test** alone).

```
import torch
from torch import nn

model = nn.Linear(n_features, 1)          # z = Wx + b
criterion = nn.BCEWithLogitsLoss()
optimizer = torch.optim.AdamW(model.parameters(), lr=3e-3)

optimizer.zero_grad()
logits = model(x).squeeze(-1)             # (B,)
loss = criterion(logits, y)                # y in {0, 1}
loss.backward()
optimizer.step()
```

Do **not** use `sklearn.linear_model.LogisticRegression` for this stone. DATA 322 already did the closed-form / sklearn version; here the point is the training loop you will reuse for the MLP.

Report:

- training-set prevalence of **serious_damage**;
- validation loss, and either ROC-AUC or Brier score;
- a confusion table at **one** probability threshold you chose;
- one sentence on why that threshold is a **decision**, not a property of z .

4. One training-loop check

Paste a few lines showing `zero_grad` $\rightarrow$ forward $\rightarrow$ `loss.backward()` $\rightarrow$ `optimizer.step()`, and state which of those produces $\nabla_{\mathbf{w}} \mathcal{L}$.

Acceptance

- Precise target language (conditional on inspection / wildfire exposure, not a premium).
- No address or APN in what you submit.
- Logistic model is PyTorch logits + `BCEWithLogitsLoss`.
- Validation numbers use the supplied split.